

Model

1 Model

1.1 Environment

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \dots, n\}$. These locations are partitioned into municipalities $g \in G$:

$$S = \bigcup_{g \in G} S^g, \quad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed supply of land $\bar{H}_i$, allocated between residential and commercial use by local zoning:

$$\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where $\bar{H}_i^R$ is residential floor space capacity and $\bar{H}_i^F$ is commercial floor space capacity. Both are chosen by the municipal government and assumed to be fully utilized in equilibrium.

Locations are connected by a transportation network. The generalized commuting cost from residence i to workplace j is

$$\tau_{ij} = \tau_{ij}(I_{ij}, n_{ij}),$$

where I_{ij} is transportation investment on link (i, j) and n_{ij} is the equilibrium commuter flow on that link. Investment reduces commuting costs and congestion raises them:

$$\frac{\partial \tau_{ij}}{\partial I_{ij}} \leq 0, \quad \frac{\partial \tau_{ij}}{\partial n_{ij}} \geq 0.$$

Following Fajgelbaum and Schaal, 2020, I parametrize the commuting cost as

$$\tau_{ij} = \delta_{ij}^\tau \frac{n_{ij}^\rho}{I_{ij}^\gamma},$$

where $\delta_{ij}^\tau > 0$ captures geographic features of link (i, j) and $\rho, \gamma > 0$.

There are three types of agents: households, competitive firms, and governments. Households choose residence and workplace locations. Firms produce a freely tradable good using labor and commercial floor space. Government decisions operate at two levels: municipal governments choose local zoning, and a metropolitan transportation authority chooses investment across all links. In the decentralized equilibrium, municipalities and the transportation authority choose policies simultaneously, each treating the other's decisions as given.

1.2 Households

A household chooses a residence location i and a workplace j to maximize utility. Preferences over the composite good Q_{ij} and residential floor space h_{ij}^R are Cobb-Douglas:

$$u_{ij}(\omega) = \frac{B_i}{\tau_{ij}} \left(\frac{Q_{ij}}{\alpha} \right)^\alpha \left(\frac{h_{ij}^R}{1-\alpha} \right)^{1-\alpha} v_{ij}(\omega),$$

where $B_i > 0$ is a residential amenity at location i , $\alpha \in (0, 1)$ is the expenditure share on the composite good, and $v_{ij}(\omega)$ is an idiosyncratic preference shock drawn independently for each (i, j) pair.

The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R \leq w_j - \phi(N_i, L_i),$$

where q_i^R is the residential rent at location i , w_j is the wage at workplace j , and $\phi(N_i, L_i)$ is a monetary disutility from residing at i . The disutility captures two sources of local negative externalities: residential congestion from neighbors and the presence of workers near the residential location:

$$\frac{\partial \phi}{\partial N_i} > 0, \quad \frac{\partial \phi}{\partial L_i} > 0,$$

where N_i denotes residential population and L_i denotes employment at location i (defined below).

Define effective income as $Y_{ij} \equiv w_j - \phi(N_i, L_i)$. Optimal demands are $Q_{ij}^* = \alpha Y_{ij}$ and $h_{ij}^{R*} = (1 - \alpha) Y_{ij} / q_i^R$. Substituting into the utility function yields the deterministic component of indirect utility:

$$\tilde{V}_{ij} = \frac{B_i Y_{ij}}{\tau_{ij} (q_i^R)^{1-\alpha}}.$$

The idiosyncratic shock $v_{ij}(\omega)$ is drawn from a Fréchet distribution with $\Pr(v_{ij} \leq z) = \exp(-T_i E_j z^{-\epsilon})$, where $T_i > 0$ and $E_j > 0$ are location-specific scale parameters and $\epsilon > 1$ governs the dispersion of idiosyncratic tastes. By the Fréchet aggregation property, the probability that a household chooses residence i and workplace j is

$$\pi_{ij} = \frac{T_i E_j \tilde{V}_{ij}^\epsilon}{\Phi}, \quad \Phi \equiv \sum_{i'} \sum_{j'} T_{i'} E_{j'} \tilde{V}_{i'j'}^\epsilon, \quad (1)$$

and expected utility is

$$U = \Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon}. \quad (2)$$

Let N denote total metropolitan population, fixed in a closed economy. Equilibrium residential population at each location is

$$N_i = N \sum_j \pi_{ij}.$$

Residential floor space market clearing at location i pins down the equilibrium rent:

$$q_i^R = \frac{(1 - \alpha) \sum_j N \pi_{ij} Y_{ij}}{\bar{H}_i^R}. \quad (3)$$

1.3 Firms

At each location j , competitive firms produce a freely tradable good using labor L_j and commercial floor space H_j^F :

$$Y_j = A_j L_j^\gamma (H_j^F)^{1-\gamma},$$

where $A_j > 0$ is location-specific productivity and $\gamma \in (0, 1)$. Taking wages w_j and commercial rents q_j^F as given, profit maximization yields:

$$w_j = \gamma A_j \left(\frac{H_j^F}{L_j} \right)^{1-\gamma}, \quad q_j^F = (1-\gamma) A_j \left(\frac{L_j}{H_j^F} \right)^\gamma.$$

Commercial floor space is constrained by zoning: $H_j^F = \bar{H}_j^F$. Labor market clearing at j gives

$$L_j = N \sum_i \pi_{ij},$$

determined by household location choices. The disutility $\phi(N_i, L_i)$ uses L_i to denote employment at location i acting as a workplace, which is this same quantity evaluated at $j = i$.

1.4 Municipal Zoning

Each municipality $g \in G$ chooses the land-use allocation $\{\bar{H}_i^R, \bar{H}_i^F\}_{i \in S^g}$ to maximize the welfare of its incumbent residents. Following the political economy of local land use, municipal governments are disproportionately responsive to incumbent landowners, who benefit from rental income but bear the costs of local congestion and worker influx. Formally, municipality g solves

$$\max_{\{\bar{H}_i^R\}_{i \in S^g}} \sum_{i \in S^g} \left[q_i^R \bar{H}_i^R + q_i^F \bar{H}_i^F - \phi(N_i, L_i) \right]$$

subject to $\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i$ for each $i \in S^g$, taking other municipalities' allocations $\{\bar{H}_{i'}^R, \bar{H}_{i'}^F\}_{i' \notin S^g}$ and transportation investment $\{I_{jk}\}$ as given.

Substituting $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$ and differentiating with respect to $\bar{H}_i^R$, the interior first-order condition at location $i \in S^g$ is

$$\underbrace{\frac{\partial(q_i^R \bar{H}_i^R)}{\partial \bar{H}_i^R} - \frac{\partial \phi}{\partial N_i} \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\text{net marginal value of residential}} = \underbrace{\frac{\partial(q_i^F \bar{H}_i^F)}{\partial \bar{H}_i^F} - \frac{\partial \phi}{\partial L_i} \frac{\partial L_i}{\partial \bar{H}_i^F}}_{\text{net marginal value of commercial}}. \quad (4)$$

The municipality equates the net marginal return to allocating the last unit of land to residential versus commercial use. Both disutility terms work in the direction of restriction: incumbents dislike additional residents (raising N_i) and additional workers (raising L_i), depressing the optimal capacity on both margins relative to a planner who does not place weight on ϕ .

1.5 Transportation Authority

The metropolitan transportation authority chooses investment $\{I_{jk}\}_{j,k \in S}$ to maximize aggregate household welfare, taking the full zoning vector $\{\bar{H}_i^R, \bar{H}_i^F\}$ as given:

$$\max_{\{I_{jk}\}} N \cdot \Gamma\left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon}$$

subject to the budget constraint

$$\sum_{j,k} \delta_{jk} I_{jk} \leq K, \quad I_{jk} \geq 0,$$

where δ_{jk} is the per-unit cost of investment on link (j, k) and K is the total infrastructure budget.

Let $\mu \geq 0$ be the Lagrange multiplier on the budget constraint. The interior first-order condition for link (j, k) is

$$\frac{N}{\epsilon} \Gamma\left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon-1} \frac{\partial \Phi}{\partial I_{jk}} = \mu \delta_{jk}. \quad (5)$$

The left-hand side is the marginal social benefit of investment on link (j, k) : lower commuting costs raise $\tilde{V}_{ij}$ for all commuters on that link, increasing Φ and hence aggregate welfare. The authority equates the marginal benefit per dollar of investment across all active links.

1.6 Equilibrium

Definition 1 (Decentralized Equilibrium). A decentralized equilibrium is a profile

$$\left(\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}_{i \in S}, \{I_{jk}^*\}, \{q_i^{R*}, q_i^{F*}, w_j^*\}_{i,j \in S} \right)$$

such that:

1. **Household optimality.** Commuting flows $\{N\pi_{ij}^*\}$ are given by the Fréchet probability formula at prices (q_i^{R*}, w_j^*) and investment I^* .
2. **Firm optimality.** Wages w_j^* and commercial rents q_j^{F*} satisfy the firm first-order conditions at $H_j^F = \bar{H}_j^{F*}$ and $L_j^* = N \sum_i \pi_{ij}^*$.
3. **Market clearing.** Residential floor space, commercial floor space, and labor markets clear at each location.
4. **Municipal Nash equilibrium.** Each municipality g maximizes its objective given $\{\bar{H}_{i'}^{R*}, \bar{H}_{i'}^{F*}\}_{i' \notin S^g}$ and I^* ; no municipality has a profitable deviation.
5. **Transportation authority optimality.** $\{I_{jk}^*\}$ maximizes aggregate welfare given $\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}$ subject to the budget constraint.

Conditions (iv) and (v) jointly constitute the simultaneous Nash equilibrium between municipalities and the transportation authority: each set of agents takes the other's choices as given and best-responds.

1.7 Social Planner

A social planner jointly chooses zoning allocations $\{\bar{H}_i^R, \bar{H}_i^F\}$ and transportation investment $\{I_{jk}\}$ to maximize aggregate household welfare:

$$\max_{\{\bar{H}_i^R, \bar{H}_i^F\}, \{I_{jk}\}} N \cdot \Gamma\left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon}$$

subject to the budget constraint $\sum_{j,k} \delta_{jk} I_{jk} \leq K$, land constraints $\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i$ for all i , and market clearing.

Comparing the planner’s first-order condition for zoning with the municipal condition in (4) reveals two positive terms that municipalities systematically ignore. First, municipalities do not internalize the benefits that looser zoning confers on non-incumbent households — both potential in-movers and residents of neighboring municipalities — who gain from lower rents and more accessible housing. Second, municipalities do not account for the transit-zoning complementarity: denser residential and commercial development near transit-accessible locations raises ridership and the marginal return to transit investment, generating a positive externality on the transportation authority’s problem. Road investment is comparatively less sensitive to local density, so this externality operates asymmetrically across infrastructure types.

Both omitted terms are positive, implying that decentralized municipalities choose zoning that is too restrictive on both margins relative to the social optimum:

$$\bar{H}_i^{R,SP} \geq \bar{H}_i^{R,DE}, \quad \bar{H}_i^{F,SP} \geq \bar{H}_i^{F,DE} \quad \text{for all } i \in S.$$

The decentralized equilibrium therefore features underinvestment in transit — not because the transportation authority errs given the zoning it observes, but because the zoning it faces is itself distorted by fragmented local governance.

References

- [1] Pablo D Fajgelbaum and Edouard Schaal. “Optimal transport networks in spatial equilibrium”. en. In: *Econometrica: Journal of the Econometric Society* 88 (4 2020), pp. 1411–1452.